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Electrochemical cells

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Introduction

Electrochemistry is the bridge between chemistry and electricity. It is a fundamental feature of chemistry that it is governed by the electrons in the molecules. Chemical reactions often involve transfer of electrons, i.e., oxidation and reduction of species and in electrochemistry this happens by means of electrodes connected to an electrical system. The energy of the electrons in transfer can be manipulated externally. One can say that electrochemistry provides a tool for our direct interference with the chemical systems. In other words, it is a viable and direct energy pathway between the macroscopic energy regime and the molecular energy regime between which energy can otherwise only be exchanged by a flow of heat.

Electrochemistry is very useful in the broad field of energy technologies as we can extract chemical energy directly as electrical energy in *batteries* and *fuel cells*. Likewise, we can store electrical energy in chemical systems by means of *rechargeable batteries* and synthesize fuels and chemicals with *electrolyzers*.

Electrochemical cells

Electrochemical cells are composed of electrodes and an electrolyte. The electrodes are electron conductors, and the electrochemical processes take place at the surface of them. The electrolyte is the substance between the electrodes which contains and conducts ions (charged molecules or charged atoms) generated or consumed in the electrochemical processes. Electrolytes can be liquid, perhaps an aqueous solution of ions, or they can be solid in the form of polymeric or ceramic membranes designed with ion conducting properties. The electrochemical processes at the electrodes either generate or consume electrons and these are conducted away from (or to) the surface where the reaction takes place. Likewise, ions formed or consumed at the surface of the electrodes, are transported to or from the electrode electrolyte interface by the ion conducting electrolyte.

Electrical current (electronic current) is passed through the electrodes. The electrochemical process at one electrode converts the electronic current into an ionic current which is then carried by ions in the electrolyte across to the other electrode where it is converted back to the electronic form. With an external circuit, the current circle is closed as it must always be. The only difference to a normal electrical circuit is that in part of the circle, the current is carried by ions instead of electrons. See Figure 1 for a general electrochemical cell and Figure 2 as the example of an electrolyzer cell converting water into hydrogen and oxygen. The electrolyzer cell shown has a proton conducting (i.e., hydrogen ion conducting) electrolyte. An electrical current is by definition considered positive, and it passes from the “plus” terminal to the “minus” terminals of a current supply or a battery. It flows from high voltage to low voltage. The electrons, which carry the current, move the other way against the direction of the current. In Figure 2, the electrical current pass from right to left. In the electron conducting parts it is carried by electrons (moving the opposite way) and in the electrolyte, it is carried by positive ions moving in the direction of the current. All the currents are *electrical currents*, and the different parts are *electronic currents* and *ionic currents*. Therefore, the electrodes must be electronic conductors, and the

electrolyte must be an ionic conductor. The electrolyte must not be an electronic conductor, because then the cell will be short circuited and the desired electrochemical process will not happen.

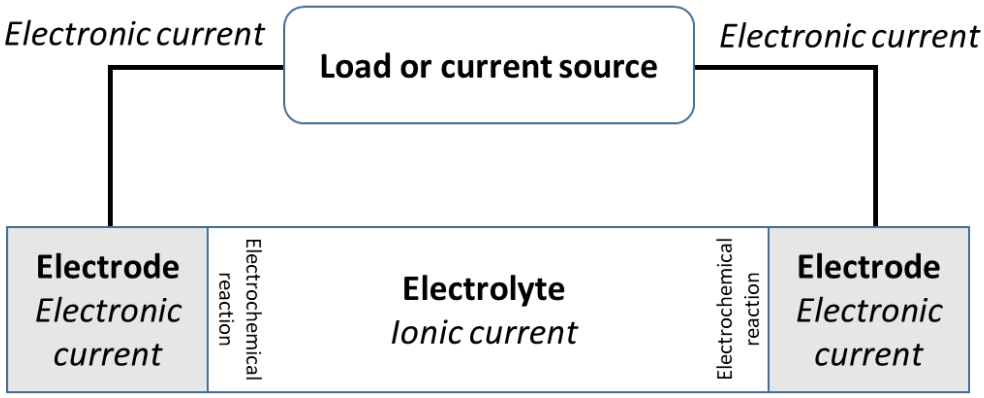


Figure 1. The electrical circuit of an electrochemical cell.

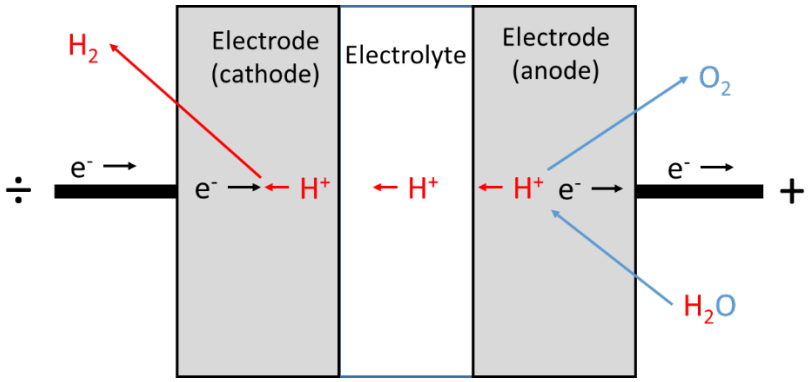


Figure 2. Schematic of a water electrolyzer cell with a proton conducting electrolyte. The current trough the cell is carried first by electrons, e^-, then protons, H^+, and finally electrons again.

Example

An example of an electrochemical cell is the lead-acid battery. We look at it during charging. At one electrode, lead sulphate ($PbSO_4$) is oxidized to lead dioxide (PbO_2) and two electrons are liberated to the electrode.

$$PbSO_4 + 2H_2O \longrightarrow PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \quad (1)$$

At the second electrode, lead sulphate is reduced to pure lead and two electrons are consumed.

$$PbSO_4 + 2e^- \rightarrow Pb + SO_4^{2-} \quad (2)$$

The electrons are exchanged via the external circuit, which is the battery charger. We also see from (1) that hydrogen ions (H^+) and sulphate ions (SO_4^{2-}) are produced at the first electrode. While the electrons are consumed by the electrode, the charge is no longer balanced in the electrolyte near the electrode since four positive charges (four H^+) are produced together with only two negative charges (one SO_4^{2-}). At the other electrode (2) only negative charge is produced (SO_4^{2-}). In order not to build up charge near the electrodes, the excess hydrogen ions in (1) and the SO_4^{2-} ions in (2) move towards the other electrode carrying jointly the ionic current across the electrolyte to maintain the charge balance.[1]

In the total reaction scheme (3), both mass and charge are balanced, and the electrons cancel out. Lead sulphate has changed to lead dioxide and metallic lead, which together contain more chemical energy than lead sulphate. The energy difference is stored in the battery.

$$2PbSO_4 + 2H_2O \rightarrow PbO_2 + Pb + 4H^+ + 2SO_4^{2-} \quad (3)$$

The difference between batteries, fuel cells and electrolyzers

Electrochemical cells can be driven by external electrical energy as in the example above. Then the current drives the process. They can also act the other way round and then the process generates the current. This is what happens when a battery is discharged or in a fuel cell. Both batteries and fuel cells generate electrical currents and are termed *galvanic cells*. Electrolyzers and batteries during charging consume electrical energy and are termed *electrolyzer cells*. The fundamental difference between batteries and fuel cells is that batteries contain all reactants and products, while fuel cells convert external reactants to external products. The dividing matrix is shown in Table 1

Table 1. The four fundamental classes of energy converting electrochemical cells.

	Galvanic cells A chemical process generates an electrical current	**Electrolyzer cells** An electrical current generates a chemical process
Internal reactants and products	Battery (during discharging)	Battery (during charging)
External reactants and products	Fuel cell	Electrolyzer

[1] One could imagine that they would combine into H_2SO_4, which is sulfuric acid, but that does not happen since they are dissolved in water. In aqueous sulfuric acid, one or both H^+ are liberated. H^+, HSO_4^-, and SO_4^{2-} coexist in chemical equilibrium.

Electrical potential, current and energy

An overview of the energy aspects of electrochemistry could start with a link between potential, current and energy. Electrical potentials are defined in a rather abstract manner. The electrical potential at a location is the work per charge necessary to bring a small charge from a place free of any electrical interaction/forces (could be outer space) and all the way to the location of interest. This is not very practical and is only meant as a concept. We cannot practically measure absolute potentials, but it doesn't matter since absolute potentials are not relevant. Like with the thermodynamic energy functions, we are only interested in differences, potential differences, and these can be measured in many cases. An electrical potential difference represents a driving force for moving a charge. When a charge moves between two potentials, it either loses or gains energy (potential energy) depending on the direction. The energy (work, *W*) absorbed or liberated by the charge is the product of the charge, *Q*, and the potential difference travelled, Δ*E*.

$$W = Q\Delta E \tag{4}$$

The charge passing pr. time unit, *t*, is the current, *I*.

$$Q = It \tag{5}$$

The current in a wire is also movement of charge over a potential difference, i.e., an electrical field through the wire. Combining (4) and (5) gives

$$W = I\Delta Et \tag{6}$$

And since power, *P*, is energy per time, then

$$P = I\Delta E \tag{7}$$

If the medium that the charge passes through has a resistance, *R* (this is practically always the case with the exceptions of superconductors and absolute vacuum), the charge is "taxed" and loses some of the energy it gained from the potential differences. If the resistance is constant and independent of the current, the energy loss can be described as a voltage loss (potential loss) by Ohms law

$$\Delta E = RI \tag{8}$$

The charge and discharge of a battery are sketched in Figure 3. The energy delivered during discharge is *IEt* or *EQ* and must be absorbed by the external load, which is a lightbulb, ⊗, in the figure. The current is controlled by the voltage of the battery and the resistance of the load. If the battery is fit for the task, it maintains the rated voltage and all the energy is consumed by the load (and a small fraction in the wire) and converted to light and heat in the light bulb. If the resistance of the load is small, the current becomes large and perhaps larger than what the battery can deliver. Then the voltage of the battery decreases and so does the current until a situation is reached where the battery can keep up. The extreme case is a shortcut where the load is practically bypassed or has near-zero resistance. Even in this case all the energy delivered by the battery must go somewhere (first law) and current and voltage will again adjust to the normally negligible, yet non-zero, resistance of the wire.

When the battery is charged, current is forced in the opposite direction and the energy *EIt* is stored in the battery by means of chemical changes inside. Seen from the outside, the battery behaves like a resistor but is changing the resistance somewhat during the charging. Some batteries show that they are fully charged by a steep increase of resistance, others don't.

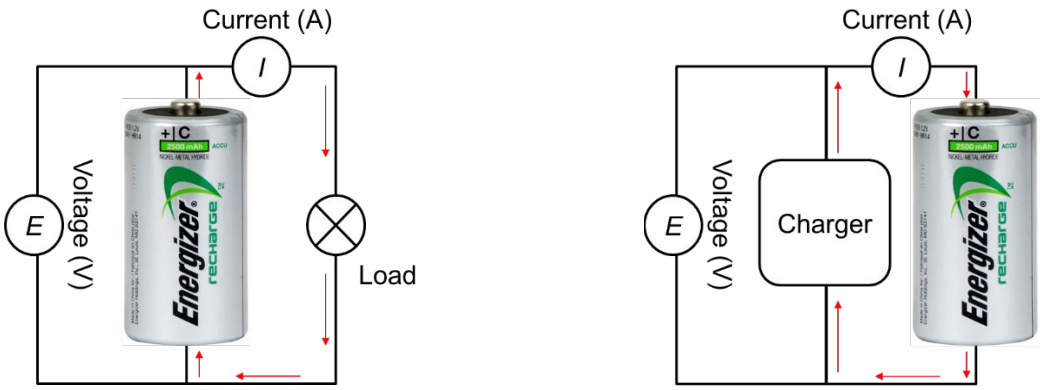


Figure 3. Discharge (left) and charge (right) of a battery.

Electrochemical reactions

There are many kinds of electrode processes. They can be divided into two main kinds, the faradaic and the non-faradaic. A faradaic process (or reaction) involves electron transfer between the electrolyte and the electrode. It causes an oxidation or a reduction of something (i.e., it increases or decreases the oxidation state of the species involved). A non-faradaic process does not involve electron transfer, but it still affects the electrode potential. The way this is possible is that charged species in the electrolyte accumulate near the electrode surface and are balanced by a similar accumulation of charge of the opposite signing in the electrode. This is called *double layer charging,* and the electrode-electrolyte interface behaves like a capacitor. We shall not deal further with non-faradaic processes; only note that they exist and are important for some diagnostics techniques (e.g., electrochemical impedance spectroscopy).

A faradaic reaction is written like any other chemical reaction, only, in addition to molecules and ions, it involves electrons, e^-. Note that the charge balance across the reaction arrow must include the charge of the free electrons. See the equations (9) and (10).

Anode and cathode

In electrochemistry, the terms *anode* and *cathode* are frequently used. The definition is as follows:

The *anode* is the electrode where an oxidation takes place. Something is oxidized and increases its oxidation state at the anode and electrons are liberated to the anode. An *anodic process* is an oxidation process, and an *anodic current* is the electron current generated form an anodic process. A simple anodic process is the oxidation hydrogen to protons (positive hydrogen ions).

$$H_2 \rightarrow 2H^+ + 2e^- \quad \text{(anodic)} \tag{9}$$

The *cathode* is the electrode where reduction takes place. Something is reduced at the cathode, and electrons for that are supplied by cathode. A *cathodic process* is a reduction process, and a *cathodic current* is the current generated form a cathodic process. The equivalent cathodic process of the one in (9) is the reduction of protons to molecular hydrogen

$$2H^+ + 2e^- \rightarrow H_2 \quad \text{(cathodic)} \tag{10}$$

The implication on fuel cells and electrolyzers is slightly inconvenient. For the fuel cell, the anode is the fuel electrode (hydrogen electrode) because hydrogen is oxidized to positive hydrogen ions and the cathode is the oxygen electrode where oxygen is reduced. So far, no problems, but when we repeat the logic with the electrolyzer, we define the anode as the oxygen electrode, because oxygen ions are now oxidized into oxygen molecules. Likewise, the cathode is now the hydrogen electrode. When fuel cell people meet electrolyzer people, this can lead to some confusion. For rechargeable batteries, it is even worse. Here the same cell changes the electrode naming when the mode is changed from charge to discharge. It is in both cases common to refer to the electrodes as the *positive* and the *negative* electrode since that does not change. For fuel cells and electrolyzers, the terms *hydrogen electrode* and *oxygen* (or air) *electrode* are also commonly used.

From thermodynamics to electrochemistry

In thermodynamics, we have established that the Gibbs free energy of a reaction quantifies the maximum work we can obtain form a chemical process, and an electrochemical process is no different. We don't need to know how to extract this work, only that it is not prevented by thermodynamics. The following derivation is based on one mole of the reaction of interest.

$$w_{max} = -\Delta g_r \tag{11}$$

Where w_{max} is the maximum molar work obtainable (form outside. Her it is practical to see the work as a positive quantity despite it leaved the system, therefore the minus before the free energy). We also know that the maximum work we can obtain from a charge moved through an electrical field, or between two electrical potentials, is the product of the charge (q) transferred and the potential difference (ΔE)

$$w_{max} = q\Delta E \tag{12}$$

Current is ordered and carries no entropy, only maximum grade energy, and can be converted completely to work. In an electrochemical cell, the potential difference is the cell voltage, E_{cell}. This holds in all cases, not only at equilibrium.

$$qE_{cell} = -\Delta g_r \tag{13}$$

The charge transferred while carrying out one mole of the reaction is the charge of one mole electrons, which is given by Faraday's constant, *F*, (*F* = 96485 C mol^{-1}) multiplied by the number of electrons in the reaction, *n* (in (9) and (10), *n* = 2), and then (13) can be rearranged into

$$E_{cell} = \frac{-\Delta g_r}{nF} \tag{14}$$

This is the ideal reversible cell voltage of a cell with index "r" referring to the *reactions* inside the cell. If the free energy of reaction is negative, the cell voltage becomes positive. In case of a battery, this is the discharging process.

The signing problem

The derivation of (14) assumed that the cell is a galvanic cell (an electrical energy producing cell), i.e., a battery during discharge or a fuel cell. E_{cell} is an equilibrium voltage and the value is irrespective of the process following. In case of a battery, E_{cell} is the same regardless of if the battery is to be charged further or discharged. However, the signing of (14) assumed discharging, but when the process is reversed, as it is during charging, the signing of Δg_r changes and if (14) is used directly, E_{cell} becomes negative, which makes little sense as a voltmeter across would read the same value no matter if the cell is meant as an electrolyzer cell or a galvanic cell.

The solution is to adopt a signing that returns a positive cell voltage[2]:

$$\begin{aligned} E_{cell} &= \tfrac{-\Delta g_r}{nF} \text{ (galvanic cell)} \\ E_{cell} &= \tfrac{\Delta g_r}{nF} \text{ (electrolyzer cell)} \end{aligned} \tag{15}$$

Faraday's law

There are several physical laws called Faraday's law, but in this connection the one for electrochemical conversion is the relevant one. It is named *Faraday's law of electrolysis*, or within electrochemical circles just *Faradays law*. It states that the amount of matter converted is proportional to the charge passed through the electrode or the cell. Intuitively, this makes sense. If hydrogen ions, H^+ are reduced to hydrogen molecules, H_2, it takes two mole electrons to produce one mole H_2 as can be easily seen from eq. (10). The law can be expressed in the following ways:

[2] In the search for a more mathematically consistent solution one could argue that signing should follow the thermodynamic convention that energy leaving the system is negative and energy entering positive. This is already reflected by the signing of Δg_r. Then one could argue that the "driving force", the cell voltage, should be signed. However, the actual (measured) cell voltage of a battery does not change its signing. It doesn't help either to sign the number of electrons, *n*, since the flow in and out in the two wires are the same.

$$n_{\mathrm{con}} = \frac{Q}{nF} \tag{16}$$

Where n_{con} is the number of moles of the molecule generated (or consumed) in the electrochemical reaction. *Q* is the number of moles elementary charges converted, and *nF* is the charge transferred for the conversion of one mole substance.

Mass based the law is

$$m_{\mathrm{con}} = \frac{MQ}{nF} \tag{17}$$

Where m_{con} is the mass converted and *M* is the molar mass of the species in focus. Finally, the law can be expressed by the electrical current, *I*, and the molar production rate of the species, $\dot{n}_{\mathrm{con}}$.

$$\dot{n}_{\mathrm{con}} = \frac{I}{nF} \tag{18}$$

Half-cells and standard potentials

Electrochemical cells always involve a pair of electrodes. Cells can only operate if the full electrical current cycle is closed. This implies that we cannot measure the voltage of one electrode only. We can only measure voltage differences (a voltmeter has two probes). Each of the two electrode processes are called *half-cell reactions* and examples are the reactions (8) and (9). One electrode can exist in electrochemical equilibrium with its electrolyte, but we can never draw or apply an external current. With only one electrode and thus only one half-cell reaction (oxidation or reduction), charge will accumulate at the electrode as electrons and in the electrolyte as ions. In a very short while, the process will stop due to the generated charge imbalance which counteracts further reaction.

Despite that half-cells always appear in pairs and that the potential of one electrode cannot be measured individually, it is convenient to assign a potential to each of them in a way that makes the potential difference between them equal to the difference in the values of the individual half-cell reactions.

$$E_{\mathrm{cell}} = E_{\mathrm{halfcell2}} - E_{\mathrm{halfcell1}} \tag{19}$$

Since such a potential depends on concentration, pressure and temperature, it is also convenient to define a standard state for the involved species. This standard state is defined like in general thermodynamics, namely by standard temperature, 298.15 K,[3] and pressure, 1 bar[4] of each species. If

[3] Strictly speaking there is no standard temperature, but 25 °C is commonly used as stated in the tables.
[4] Standard pressure is generally 1 bar as recommended by IUPAC since 1982, but since many accurate tables were generated based on the previous standard pressure of 1 atm (1.01325 bar), this is still valid for standard potentials. The difference is negligibly small since it is the effect of only a 1% pressure change.

dissolved species are involved the concentration is 1 molar (1 M). The potential is then the standard potential, E^0.

The problem of measuring half-cell potentials is circumvented by defining one electrode reaction as reference and measure all the rest with respect to this reference. For this, the hydrogen process in (20) is chosen. The process is also that of the *standard hydrogen electrode* (SHE).

$$2H^+ + 2e^- \rightleftarrows H_2 \qquad (20)$$

All species must be in the standard state with *activity* 1 and the solvent must be water. Activity 1 is achieved by H^+ at 1 M (pH = 0) and hydrogen at 1 bar. This definition is valid at any temperature. A brief explanation of activity can be found in the following box and

Table 2 guides the transcription of activity into pressure or concentration, which is more practical. Note that pure liquids and pure solids have the activity 1 by definition.

A note on activity, *a*:

Activity is a concept in thermodynamics used to quantify substances instead of using partial pressure of gasses or concentration species in a solution. Pressure and concentration are more intuitive but suffer from the fact that their actual chemical impact is somewhat different from what it should be at a given pressure or concentration. This is due to non-ideal behaviour and the shielding effects of other molecules. It may be that the chemical equilibrium established in a real system is slightly different from what it should be with the given concentrations and/or pressures because some species are "seen" as if the concentration is lower than it is. Activity mitigates that by taking the value that makes equilibrium and concentration (or pressure) fit. Activity is then linked to concertation or pressure by activity coefficients that can be estimated by different more or less complicated tricks. The activity of pure condensed phases (pure liquids and solids) is by definition 1.

Thus, accurate calculation can be complicated, and it is common practice simply to replace activities with partial pressures or concentrations. The error is often quite small and acceptable. Refer to

Table 2.

A nice feature of activity is that it is dimensionless. This is convenient when it is used in the Nernst equation, which is introduced later.

Table 2. Activities of various species. Pressure and concentration are divided by the standard pressure, p^0, and the standard concentration, c^0, respectively. This way activity becomes dimensionless and errors due to the use of alternative units for pressure and concentration are avoided.

Species	**Characterized by**	**Activity**	**Note**
Pure gas	Pressure, p	$\frac{p}{p^0}$	p^0 = 1 bar
Gas component, i, in gas mixture	Partial pressure, p_i	$\frac{p_i}{p^0}$ (approximated)	p^0 = 1 bar
Pure liquid	-	1	
Species, i, dissolved in liquid	Concentration, c_i	$\frac{c_i}{c^0}$ (approximated)	c^0 = 1 M (mol L^{-1})
Pure solid	-	1	

Note: the symbols p^0 and $p^\ominus$ are both used for the standard pressure. It may refer to 1 atm. in older references.

After the reference reaction (20) has been defined, the electrode potential of any other electrode reaction can be measured. If all involved species of the second electrode system are at unit activity like in the SHE then the potentials are standard potentials. Table 3 contains a limited selection of standard potentials. Note that the standard potential in general vary with temperature. It is only the value for (20) that is fixed for all temperatures by definition.

Table 3. Standard potentials (E^o) for a small selection of electrode processes at 25 °C.

Reaction	E^o **(V)**	**Reaction**	E^o **(V)**
$Li^+ + e^- \rightleftarrows Li$	-3.01	$2H^+ + 2e^- \rightleftarrows H_2$	0.0000
$Ca^{2+} + 2e^- \rightleftarrows Ca$	-2.84	$Cu^{2+} + 2e^- \rightleftarrows Cu$	0.34
$Na^+ + e^- \rightleftarrows Na$	-2.71	$Cu^+ + e^- \rightleftarrows Cu$	0.52
$Mg^{2+} + 2e^- \rightleftarrows Mg$	-2.84	$Hg^{2+} + 2e^- \rightleftarrows Hg$	0.80
$Al^{3+} + 3e^- \rightleftarrows Al$	-1.66	$Ag^+ + e^- \rightleftarrows Ag$	0.80
$Zn^{2+} + 2e^- \rightleftarrows Zn$	-0.76	$Br_2 + 2e^- \rightleftarrows 2Br^-$	1.07
$Fe^{2+} + 2e^- \rightleftarrows Fe$	-0.44	$O_2 + 4H^+ + 4e^- \rightleftarrows 4H_2O$	1.23
$Cd^{2+} + 2e^- \rightleftarrows Cd$	-0.40	$Cl_2 + 2e^- \rightleftarrows 2Cl^-$	1.36
$Ni^{2+} + 2e^- \rightleftarrows Ni$	-0.23	$F_2 + 2e^- \rightleftarrows 2F^-$	2.87

The standard cell voltage of an electrochemical cell can be calculated as the difference of the standard potentials of the reactions taking place on the two electrodes. If a battery was to be constructed from the Zn reaction and the Ag reaction from Table 3, then the standard cell voltage would be

$Ag^+ + e^- \rightleftarrows Ag$	0.80 V
$Zn^{2+} + 2e^- \rightleftarrows Zn$	-0.76 V
Total difference	1.56 V

For an acidic hydrogen fuel cell or a water electrolyzer at 25 °C the calculation is

$O_2 + 4H^+ + 4e^- \rightleftarrows 2H_2O$	1.23 V
$2H^+ + 2e^- \rightleftarrows H_2$	0.00 V
Total difference	1.23 V

This holds at 25 °C and provided that all activities are 1, which is true for the two metals as they are pure condensed phases. The ions and gasses will need to have unit activities (1 M and 1 bar). For other activities, refer to the Nernst equation, which is introduced below. For other temperatures, the standard potentials need to be corrected (except for that of hydrogen, (20), which is defined as zero at any temperature).

Standard cell potential calculated from chemical reactions

The *standard cell voltage* of an electrochemical cell can be calculated from the standard Gibbs free energy of the total reaction by (14), and the standard Gibbs free energy of reaction can be found from the change in standard Gibbs free energy of formation of the involved species.

Variation of equilibrium cell voltage with pressure, concentration, and temperature

The Nernst equation – variation with pressure and concentration

The standard potential of an electrochemical cell only applies if all reacting species and products are at unit activity, or with good approximation at 1 bar or 1 molar concentration. This is often not the case and the tool to calculate electrode potentials with the reactants at other pressures and concentration is the *Nernst equation*. Equilibrium is still assumed but now at other pressures and concentrations.

The basic idea is to keep the standard cell voltage, E^0_{cell}, at the chosen temperature and then add a compensating term to quantify the deviation from the standard value.

$$E_{\text{cell}} = E^0_{\text{cell}} - \text{compensation} \tag{21}$$

The compensation is a function of the activity of all reactants and products of the overall process. Let's take the reaction of a hydrogen powered fuel cells as an example:

$$2H_2 + O_2 \rightleftarrows 2H_2O \tag{22}$$

Then the equilibrium cell voltage, E_{cell} is

$$E_{cell} = E^0_{cell} - \frac{RT}{nF}\ln\frac{a^2_{H2O}}{a_{O2}\ a^2_{H2}} \tag{23}$$

This is the *Nernst equation* for a hydrogen powered fuel cell. *a* is the activity of a species, se the box on activity on page 10. The Nernst equation is justified in the box on page 14. We shall replace activities with pressures shortly.

The logarithmic form of the compensation term in (24) secures that the compensation is zero if the activities of all species are 1. We see that the activities of the products are in the numerator of the argument to the logarithmic function and that the reactants are in the denominator. Each activity is raised to the power of the stoichiometric coefficient of that species. It makes sense intuitively that if the activity of the reactants (H_2 and O_2) increase, the process is promoted and the voltage of the produced electrical energy increases since the subtracted compensation term gets smaller. Likewise, if the activity of the product (H_2O) increases, the equilibrium cell voltage decreases because the reverse process is promoted.

The general expression of the Nernst equation is

$$E_{\text{cell}} = E^0_{\text{cell}} \pm \frac{RT}{nF}\ln\frac{\prod_{\text{Prod.}} a_{\text{i}}^{\nu_{\text{i}}}}{\prod_{\text{React..}} a_{\text{i}}^{\nu_{\text{i}}}} \tag{24}$$

with the signing of the compensation term following the signing of Δg_r i.e., "+" in case of an electrolyzer cell (an electricity consuming cell) and "−" in case of a galvanic cell (an electricity generating cell), as discussed on page 8 (The signing problem). $\prod a_i^{\nu_i}$ means the product of all elements, $a_i^{\nu_i}$, like $\sum x_i$ means a sum. It can seem a bit arbitrary to just change the signing whenever convenient, but if that is not done, then the cell voltage of a battery would change signing when the operational mode is changed between charging and discharging.

When working with practical systems, like fuel cells, we characterize the gases by their pressure, and the activities in (23) can be replaced by pressures after $a_i = \frac{p_i}{p^0}$ as listed in

Table 2. For simplicity p^0 is often omitted in the writing despite it formally introduces a problem with the units if they don't cancel out (you cannot take the logarithm to a unit). Eq. (23) then becomes

$$E_{cell} = E^0_{cell} - \frac{RT}{nF}\ln\frac{p^2_{H2O}}{p_{O2}\ p^2_{H2}} \quad (\textit{all pressures in bar without the unit}) \tag{25}$$

If water is generated as pure liquid water, then the activity is 1 as for all pure condensed phases and (25) simplifies to (26)

$$E_{cell} = E^0_{cell} - \frac{RT}{nF}\ln\frac{1}{p_{O2}\ p^2_{H2}} \tag{26}$$

or even simpler

$$E_{cell} = E^0_{cell} + \frac{RT}{nF}\ln\left(p_{O2}\ p^2_{H2}\right) \tag{27}$$

If water is liberated as vapour, its partial pressure must be considered and (25) applies. Often, water is assumed to be liquid in fuel cells although this assumption is not always well justified.

Eq. (25) to (27) are valid for both fuel cells and electrolyzers, because p_{O2} and p^2_{H2} have the same effect in the equations. It is only when the Nernst equation is set up the first time that one must consider reactants and products. After that, the equation describes the same equilibrium voltage for fuel cells and electrolyzers. The step from (26) to (27) can be explained purely mathematically where inversion of the fraction causes the signing to change, but it can also be explained electrochemically as a change of the roles of H_2 and O_2 from reactants to products.

Justification of the Nernst equation

The equilibrium cell voltage is governed by the free energy of the reaction, Δg_r, of the overall cell reaction. The standard Gibbs free energy change of the reaction is calculated from standard free energies of formation:

$$\Delta g^0_r = \sum_{product} v_i \Delta g^0_{f,i} - \sum_{reactants} v_i \Delta g^0_{f,i} \tag{28}$$

A similar equation for the free energy of reaction with the species not at standard state, Δg_r , i.e., at different pressures and concentrations can be expressed as

$$\Delta g_r = \sum_{product} v_i g_i - \sum_{reactants} v_i g_i \tag{29}$$

where g_i is the free energy of each of the involved species derived from free energies of formation. Thermodynamics states elsewhere that the pressure dependence on the free energy of a perfect gas can be given by

$$g_i = \Delta g^0_{f,i} + RT\ln a_i \tag{30}$$

Where a_i is the activity of species i. If **Error! Reference source not found.** is inserted into (29), then we get

$$\Delta g_r = \sum_{product} \left(v_i \Delta g^0_{f,i} + v_i RTln\ a_i\right) - \sum_{reactants} \left(v_i \Delta g^0_{f,i} + v_i RTln\ a_i\right) \tag{31}$$

which rearranges into

$$\Delta g_r = \left[\sum_{product} v_i \Delta g^0_{f,i} - \sum_{reactants} v_i \Delta g^0_{f,i}\right] + \left[\sum_{product} v_i RT ln\, a_i - \sum_{reactants} v_i RT ln\, a_i\right] \quad (32)$$

With first bracket rewritten in accordance with (28) and the second bracket contracted:

$$\Delta g_r = \Delta g^0_r + RT ln \frac{\prod_{\text{Prod.}} a_{\text{i}}^{\nu_{\text{i}}}}{\prod_{\text{React..}} a_{\text{i}}^{\nu_{\text{i}}}} \quad (33)$$

The cell voltage is obtained by division of (33) by the charge transferred, *nF*.

$$E_{\text{cell}} = E^0_{\text{cell}} + \frac{RT}{nF} \ln \frac{\prod_{\text{Prod.}} a_{\text{i}}^{\nu_{\text{i}}}}{\prod_{\text{React..}} a_{\text{i}}^{\nu_{\text{i}}}} \quad (34)$$

In eq. (34) no measures are taken to avoid negative cell voltages. To do that, the signing of the compensation term (left-most term) must be decided case to case as discussed on page 8 (The signing problem) and we get (35), the same as (24):

$$E_{\text{cell}} = E^0_{\text{cell}} \pm \frac{RT}{nF} \ln \frac{\prod_{\text{Prod.}} a_{\text{i}}^{\nu_{\text{i}}}}{\prod_{\text{React..}} a_{\text{i}}^{\nu_{\text{i}}}} \quad (35)$$

Activities can then be approximated to partial pressures or concentrations.

Variation with temperature

Temperature is part of the Nernst equation, but one cannot just calculate the temperature dependence by simply varying *T*. The reason is that the standard Gibbs free energies on which the standard potentials are based, also vary with temperature. The entropy, that the Gibbs free energy depends on, depends strongly on temperature. Such calculations are more complicated, and it is recommended to use tabulated values of either the standard free energy or the standard potentials at other temperatures, if possible.

Electrokinetics

Thermodynamics versus kinetics

Kinetics is the science of the rate of reaction. Reaction rates are affected by catalysts, which are other molecules or surfaces that take part in the reactions without being consumed. An example is a platinum surface on which hydrogen oxidation and reduction take place fast. Hydrogen binds to the surface during the reaction but renders it unaffected afterwards. A good catalyst can accelerate the reaction very much, but it does not change the thermodynamics and the chemical or electrochemical equilibrium, it only leads to a faster establishment of the same equilibrium.

In brief, thermodynamics decides the maximum available (or necessary) energy of a process, but not the rate of it. Kinetics sets the rate of reaction, i.e., the power in energy systems. Poor (slow) kinetics leads to imperfections and losses and not all the maximum available energy given by thermodynamics is utilized in real systems. Charging a battery requires more energy than what can be reclaimed during discharge, and the difference in energy is wasted as heat. Thermodynamics does not foresee that and cannot prevent it either.

Polarization

Polarization means that the potential of an electrode is different from that attained when the electrode is at equilibrium with its electrolyte. In an electrolyzer cell or a battery in charging mode, we can speed up the process by turning up the voltage, but we still convert the same amount of substance per mole electrons transferred because that is governed by Faraday's law, (18), which depends on current and not voltage. Since the power consumed is voltage times current, higher voltage at a given current means higher energy spent to convert the amount of stuff given by the current. Polarization accelerates the process since voltage is the driving force. The penalty is lower energy efficiency.

If the cell is a galvanic cell that produces electrical energy, part of the voltage generated us used on polarization to promote the current. Again, the penalty is lower energy efficiency.

Overvoltage

The additional voltage needed to promote an electrode process is called an *overvoltage*. It does not matter if this leads to higher or lower voltages (we do not talk about *undervoltages*). What matters is that the voltage is shifted away from what is expected from idealized thermodynamics, which only treats equilibria and reversible processes. A reversible process is infinitively slow and since we cannot wait for that in practical applications, we change the potential a bit to force the process during charging and we accept a slightly lower cell voltage during discharge. A common principle is that the overvoltages are always working against the purpose of the electrochemical cell (with a few exceptions that are irrelevant in this context). When we want to drive a process by electrical energy, the voltage is the "driver", and we need to apply a higher voltage. When we like a cell to generate electrical energy, we pay a penalty via a lower voltage.

The physical origin of overvoltages are

- Voltage loss due to the ohmic resistance in the electrolyte and in cables and connections. This loss is purely ohmic, i.e., it follows Ohms law, which dictates that the voltage loss is proportional to the current.
- Electrochemical losses at the electrodes. The charge transfer process can be slow and by polarizing the electrode (changing its potential) electron transfer in one direction is favored and the process is accelerated. This contribution is not at all ohmic but follows a logarithmic like characteristic with current. This overvoltage is referred to as *charge transfer overvoltage*, *kinetic overvoltage* or simply *anodic or cathodic overvoltage*.

- At high currents, transport of species to and from the reaction sites on the electrodes can be a limiting factor. This may in some cases result in *transport overvoltages*, which are also called *concentration overvoltages*.

Overvoltages are also called *overpotentials*. They apply to batteries, fuel cells and electrolyzers the same way.

Electrocatalysts

The rate of electrochemical processes depends heavily on the quality of the reaction sites on the electrode. I most cases an electrocatalyst is applied there. An electrocatalyst is a material that is optimized for promoting the charge transfer reaction. Like other catalysts it takes part in the process but is left unchanged afterwards and thus not consumed. A proper choice of catalyst is often more effective than a high polarization, but both a catalyst and some degree of polarization are mandatory in practical electrochemical cells for conversion of energy. Typical catalyst materials are metallic nanoparticles, noble metals, and a variety of metal oxides.

Efficiencies of electrochemical cells

There are different efficiencies defined for electrochemical cells.

Faradaic efficiency (current efficiency)

Faradaic efficiency is the degree of fulfilment of Faraday's law. The efficiency of conversion of the fuel molecules to electrons (galvanic cell, battery) or electrons to product molecules (electrolysis cell, battery).

Theoretically, the amount of substance converted in the electrochemical processes should exactly match the charge passed through the cell. In real systems, though, this may not always be the case. Parasitic processes may consume some charge, and crossover of products through the electrolyte leads to waste of the product. Overcharging of a battery is another example as current passes without contributing to further charging. Yet, in most cases the faradaic efficiency is close to 100%.

Voltage efficiency

The ratio between the actual cell voltage and the theoretical cell voltage given by thermodynamics. The treatment is different for batteries as compared to fuel cells and electrolyzers.

Batteries

For batteries, the theoretical cell voltage is given by (15), i.e., governed by the free energy of the reaction. Since the applied voltage is slightly higher during charge and slightly lower during discharge the voltage efficiency is as a first approximation

$$\eta_{V,ch} = \frac{E_{rev}}{E_{ch}} < 100\% \tag{36}$$

and

$$\eta_{V,disch} = \frac{E_{disch}}{E_{rev}} < 100\% \tag{37}$$

$\eta_{V,ch}$ and $\eta_{V,disch}$ are the voltage efficiencies during charge and discharge, respectively. E_{ch} and E_{disch} are cell voltages during charge and discharge, and E_{rev} is the equilibrium reversible cell voltage determined by thermodynamics.

However, the voltage of a battery always changes during the charge or discharge process and therefore the voltage efficiency changes accordingly. A full constant current charging or discharging step has the voltage efficiencies in (38) and (39) based on the average applied voltage.

$$\eta_{V,ch} = \frac{E_{rev}}{\frac{1}{t}\int_{t=0}^{t} E_{ch}\,dt} < 100\% \tag{38}$$

and

$$\eta_{V,disch} = \frac{\frac{1}{t}\int_{t=0}^{t} E_{disch}\,dt}{E_{rev}} < 100\% \tag{39}$$

Here current was assumed constant during charging or discharging. This is not the case during intelligent charging, e.g., of vehicular batteries, and then the integration must be on the charge instead, but this will not be pursued here. Voltage efficiencies of batteries are typically close to 100%, say 80-95%

Fuel cells and electrolyzers

Fuel cells and electrolyzers are normally operated at constant current and voltage because they don't change during the process like a battery does so, voltage integration is not necessary.

Another difference is that the reference voltage is not the reversible voltage derived from the free energy of the reaction, but the reversible voltage derived from the enthalpy of reaction. This is called the *thermoneutral voltage*, E_{TN}, and is introduced here without further treatment. The reason for using the thermoneutral voltage for efficiency calculation is that it relates to the total chemical energy of the fuel, not only the free energy, and the energy of a fuel is its heating value, which is numerically equal to the enthalpy of reaction.

$$E_{\mathrm{TN}} = \frac{|\Delta H_{\mathrm{r}}|}{nF} = 1.48\ \mathrm{V} \tag{40}$$

On that background the voltage efficiency of fuel cells and electrolyzer cells, $\eta_{V,EC}$ and $\eta_{V,FC}$, respectively, become

$$\eta_{V,EC} = \frac{E_{TN}}{E_{actual}} \tag{41}$$

and

$$\eta_{V,FC} = \frac{E_{actual}}{E_{TN}} \tag{42}$$

For fuel cells, the voltage efficiency is normally moderate due to overvoltages, say 40-60%. For electrolyzers the voltage efficiency is higher despite electrolyzers have similar overvoltages. This is because the reference voltage is the thermoneutral voltage, which is higher than the reversible cell voltage so, higher voltage losses can be accommodated before the voltage efficiency drops significantly below 100%. Typical voltage efficiencies of electrolyzers are 70-85%, and high temperature electrolyzers show practical voltage efficiencies around 100%.

System efficiency

System efficiency is the energy efficiency of the entire system including balance of plant (the system around the electrochemical cells). End of the day this is the key performance indicator to the user.